DecodeX 2026
Round 2: Business Case Study

Supplementary Notes

Industry Context Exhibits

(For Interpretation and Managerial Context Only)

Important Note to Participants

The exhibits below are provided to support contextual understanding of India's electric mobility and urban transportation environment. They are **not additional datasets** and should **not be merged, extrapolated, or used for numerical estimation** alongside the VoltRide dataset.

All quantitative analysis must be based solely on the data explicitly provided in the case.

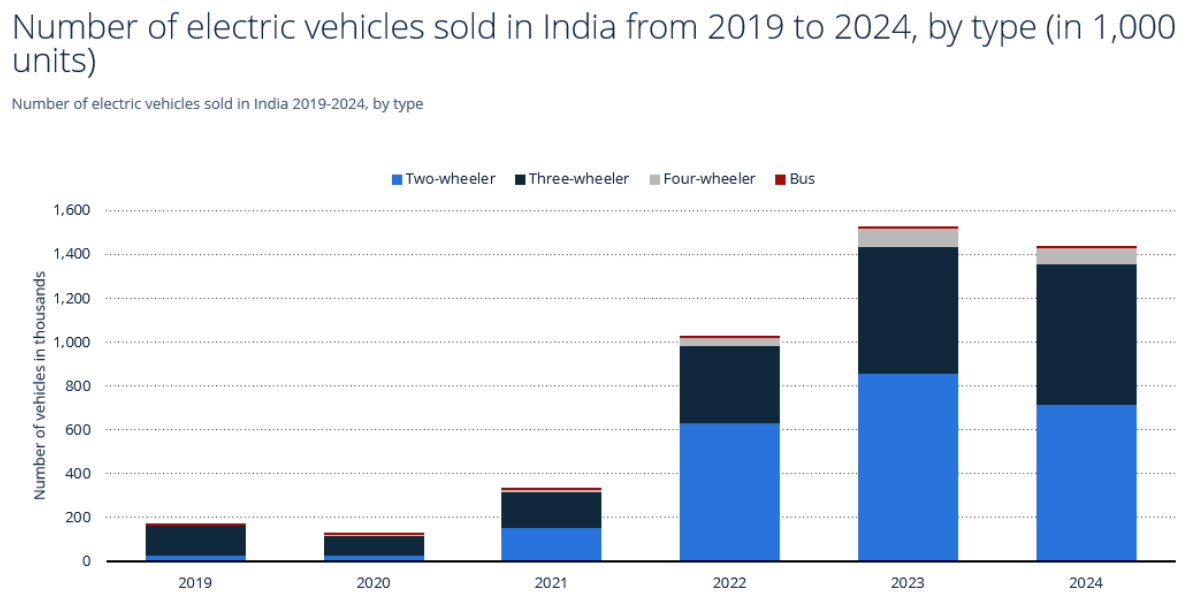
Participants should use these exhibits to:

- Anchor **managerial reasoning**
- Assess **feasibility and realism** of recommendations
- Contextualize trade-offs identified in the data

Participants should **not**:

- Extract numerical inputs
- Perform calculations using these exhibits
- Treat them as extensions of the analytical dataset

Exhibit A: India's Automobile and Electric Mobility Landscape



Source(s): Climate Trends; Ministry of Road Transport and Highways (India)

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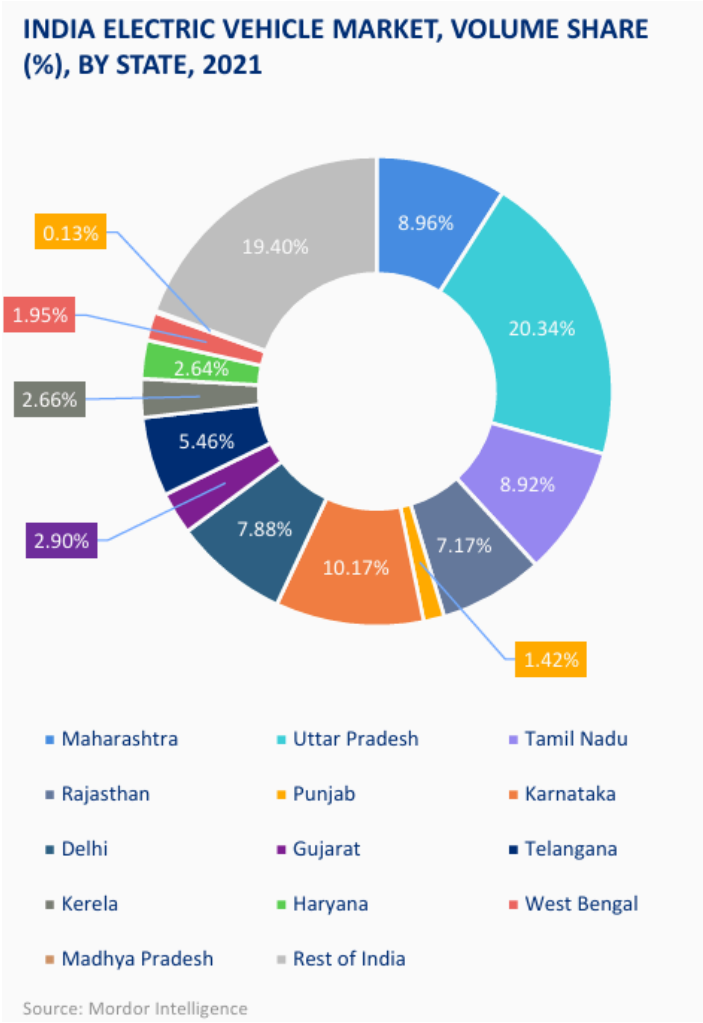


Supplementary Notes

ELECTRIC VEHICLE MARKET, VOLUME IN THOUSAND UNITS, BY STATE, **INDIA,** 2018-2027

BY STATE	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	CAGR(%)
Maharashtra	40.36	78.33	30.68	68.50	91.76	125.93	178.06	258.92	354.39	477.57	39.08
Uttar Pradesh	474.39	232.97	139.49	155.41	204.99	277.04	385.73	552.33	744.44	987.87	36.96
Tamil Nadu	11.61	14.38	25.43	68.14	90.58	123.36	173.08	249.74	339.21	453.60	38.02
Rajasthan	41.57	27.66	24.99	54.79	72.76	98.99	138.76	200.02	271.40	362.57	37.88
Punjab	3.21	4.01	3.70	10.84	14.49	19.86	28.04	40.70	55.62	74.84	38.87
Karnataka	68.14	25.81	42.33	77.70	103.59	141.50	199.12	288.18	392.58	526.53	38.43
Delhi	185.44	95.91	53.29	60.21	79.46	107.44	149.67	214.42	289.15	383.89	37.03
Gujarat	4.27	3.69	4.80	22.20	29.68	40.67	57.40	83.31	113.83	153.13	38.84
Telangana	30.39	20.38	21.85	41.70	55.36	75.30	105.53	152.09	206.33	275.58	37.85
Kerala	2.10	1.93	5.90	20.33	27.08	36.97	52.00	75.21	102.39	137.25	38.35
Haryana	40.38	21.23	13.08	20.21	26.56	35.75	49.58	70.72	94.95	125.50	36.42
West Bengal	59.92	56.23	44.93	14.94	20.00	27.43	38.76	56.32	77.05	103.77	39.00
Madhya Pradesh	1.14	1.27	0.99	0.97	1.32	1.82	2.60	3.82	5.28	7.18	40.32
Rest of India	201.44	86.74	118.79	148.30	191.44	253.04	344.25	481.27	632.67	818.02	33.70

India represents one of the world's largest and most structurally distinct mobility markets. Urban transportation demand is characterized by **high trip frequency, short average distances, and extreme peak-hour compression**. Two-wheelers and three-wheelers dominate overall vehicle volumes, while passenger cars play a disproportionate role in app-based ride-hailing.

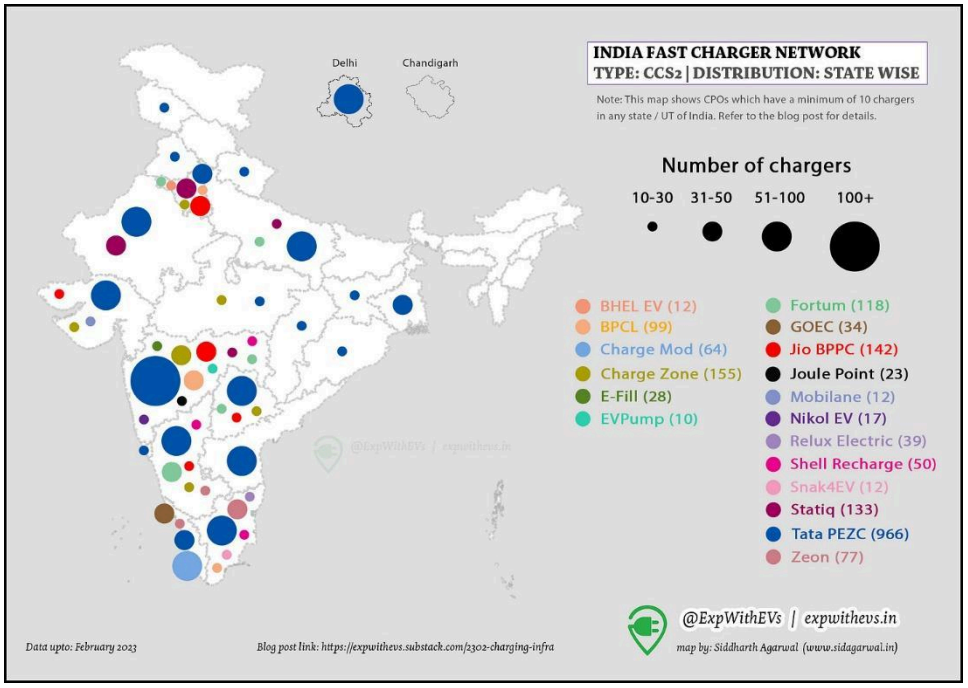
Industry reports indicate that passenger vehicle growth has been strongest in large metros, where ride-hailing increasingly substitutes for personal vehicle ownership rather than complementing public transport. EV adoption has accelerated meaningfully, but remains **segment-skewed**, with four-wheeler commercial EV fleets still facing higher operational complexity.

Managerial implication:

Assumptions drawn from private EV ownership or global markets do not directly translate to Indian EV ride-hailing operations.

Exhibit B: Charging Infrastructure Growth and Structural Imbalances

Supplementary Notes



Map of CPOs and their presence in India. Minimum of 10 chargers / state / CPO considered.

From <https://www.expwithevs.in/p/2302-charging-infra>

India's electric vehicle (EV) transition remains at an **early but accelerating stage**. As of FY23/24, India's total EV fleet—including hybrid and battery electric vehicles—was estimated at **just over 184,000 units**, representing **approximately 0.3% of the country's total vehicle fleet**. Despite this low penetration, momentum is building rapidly, supported by policy incentives, declining total cost of ownership, and growing acceptance in specific vehicle segments.

Supplementary Notes

Charging Infrastructure: Rapid Growth, Structural Gaps

India's public EV charging infrastructure expanded **nearly ninefold** in just over two years—from **1,800 stations in February 2022 to 16,347 by March 2024**. While this growth is significant, it remains **well below future requirements**. According to projections by **Forvis Mazars**, India could have **around 50 million EVs by 2030**, requiring approximately **1.3 million public charging stations**. Meeting this demand would necessitate the installation of **roughly 400,000 chargers per year**, a scale-up that presents major execution and coordination challenges.

The Ministry of Heavy Industries has taken initial steps by approving **2,877 public charging stations across states**, along with **1,576 charging stations along 16 highways and nine expressways**. However, the pace of deployment must accelerate substantially to keep up with projected EV adoption.

Market Growth Outlook and Segment Dynamics

India's EV market is projected to reach **USD 48.6 billion by 2030**, driven primarily by **two-wheelers (2W) and three-wheelers (3W)**, which dominate urban and last-mile mobility. These segments rely largely on **AC slow charging and battery swapping**, shaping the nature of charging infrastructure investments. In contrast, **four-wheelers (4W) and electric buses** require a mix of **AC and DC fast-charging solutions**, adding complexity to network planning and capital allocation.

New model launches, improved vehicle performance, falling battery costs, expanding supplier ecosystems, and policy incentives—most notably **FAME II and its expected successor (Phase III)**—are strengthening the value proposition for EV adoption across segments.

Infrastructure Constraints and Adoption Trade-offs

The dominance of 2Ws and 3Ws in India's EV mix introduces a structural trade-off. While these segments enable rapid electrification at lower cost, they **slow the development of fast-charging infrastructure** required for broader passenger and commercial EV adoption. As a result, over the **2025–2034 period**, India's total EV fleet (commercial and passenger vehicles) is expected to grow to **approximately 2.9 million units**, up from under 200,000 units in 2023—still modest relative to long-term potential. A significant portion of this growth is expected to come from **hybrid vehicles**, although there remains **strong upside risk for passenger EVs** in the medium to long term.

Policy Direction and Strategic Implications

Under the second phase of the FAME scheme and the anticipated **FAME III rollout**, the government has set a target of **500,000 public charging points by 2030**. Industry leaders, including Forvis Mazars' Transport and Logistics practice, emphasise that **scaling charging infrastructure is critical** to reducing range anxiety and unlocking mass adoption. By 2030, EVs are projected to account for **nearly one-third of India's passenger vehicle market**, underscoring the urgency of coordinated infrastructure expansion, grid readiness, and private-sector participation.

Supplementary Notes

India's public EV charging network has expanded rapidly, yet **deployment has been uneven**. Charging stations are concentrated along select corridors and commercial districts, and the **charger-speed mix** reflects historical demand from two-wheelers, three-wheelers, and depots rather than continuous four-wheeler ride-hailing usage.

Queueing and waiting times remain common during peak hours, particularly in dense urban zones. Industry projections suggest that infrastructure must scale faster to prevent persistent congestion as four-wheeler EV adoption grows.

Managerial implication:

Charging constraints are shaped by **timing, speed, and utilization behavior**, not just charger counts.

Exhibit C: Urban Demand Patterns in Indian Metros

Urban mobility demand in Indian metros is defined less by daily averages and more by **short, intense demand spikes**. Office commuting, transit nodes, and weather events (notably rainfall) drive localized surges.

Operational stress often emerges at the **zone–hour level**, where vehicle readiness fails to match demand precisely.

Managerial implication:

City-level averages mask operational risk; **micro-mismatches** drive cancellations and wait times.

Supplementary Notes

Exhibit D: EV Fleet Operations vs Conventional Ride-Hailing

Comparison of EV vs. Conventional Ride-Hailing

Feature	EV Fleet Operations	Conventional Ride-Hailing
Operational Cost	**₹1.0 – ₹1.5 per km**	**₹5.20 (CNG) – ₹6.50 (Diesel) per km**
Upfront Cost	High (₹17.5 – ₹22 Lakh)	Low (₹5.8 – ₹7.6 Lakh)
Maintenance	**40–50% lower** due to fewer moving parts	Regular servicing needed (oil, filters, etc.)
Ownership Model	Asset-heavy (company-owned)	Asset-light (driver-owned/gig model)
Pricing	Predictable; **Zero surge pricing**	Dynamic; Frequent **surge pricing**
User Experience	Reliable; Zero cancellations	Higher cancellation rates (~27%)

Unlike fuel refueling, EV charging requires extended downtime and is sensitive to congestion and driver behavior. For fleet operations, this creates a critical distinction:

- **Registered vehicles ≠ dispatchable supply**
- **Online drivers ≠ charge-ready vehicles**

Range anxiety and anticipated queueing often cause vehicles to exit service pre-emptively, amplifying shortages during peak demand.

Managerial implication:

EV ride-hailing requires **anticipatory coordination**, not reactive dispatch alone.